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Open
Geospatial
Consortium

OGC INDOORGML 2.0 PART 2B – JSON ENCODING

STANDARD
Implementation

CANDIDATE SWG DRAFT

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ABSTRACT

The IndoorGML 2.0 Part 2b – JSON Encoding Standard presents the implementation-dependent JavaScript Object Notation (JSON) encoding of the concepts defined by the IndoorGML 2.0 Part 1 – Conceptual Model Standard (OGC 22-045r5). The JSON Encoding is compliant to JSON schema specified by RFC8259. The IndoorGML JSON (IndoorJSON) encoding defines a concrete implementation schema for representing indoor spatial information, including primal space, dual space, thematic layers, interlayer connections, and navigation-related indoor features. This encoding can be used to store and exchange 2D and/or 3D indoor space and indoor navigation network models in the JSON format as data sets or via web services.

This Part 2b of the Standard defines how the concepts developed in Part 1 are realized using JSON schemas. The IndoorJSON Encoding represents a full mapping of the Conceptual Model and is derived fully automatically from the UML model following the UML-to-JSON encoding rules as defined by OGC 24-017r1. The IndoorJSON adopts JSON Schema as its primary schema language and uses geometry objects compatible with OGC Features and Geometries JSON (JSON-FG, OGC 21-045r1), where appropriate. This design choice addresses the limitation that conventional GeoJSON does not provide sufficient support for indoor 3D geometry, especially solid and volumetric geometry. Table 1 maps requirements classes from the IndoorGML 2.0 Conceptual Model into the implementation details documented in this standard.

Table 1

CONCEPTUAL MODEL	SECTION	JSON SCHEMA
IndoorGML Core	Clause 7	Annex B.1
Navigation Extension	Clause 8	Annex B.2



KEYWORDS

The following are keywords to be used by search engines and document catalogues.

OGCDoc, OGC documents, indoor, navigation, IndoorGML, JSON, GeoJSON, JSON-FG



PREFACE

In order to achieve consensus on the modeling indoor spaces and their neighborhood relationships to support indoor location-based services, a UML Conceptual Model, IndoorGML 2.0 Part 1, was approved as an OGC Standard (OGC 22-045r5) in June, 2025. This model covers geometric and semantic properties of indoor spaces relevant for indoor navigation. This IndoorGML 2.0 Part 2: JSON Encoding (IndoorJSON) Standard defines how those concepts should be realized using JSON format. The IndoorJSON encoding is intended for software developers, data providers, standards implementers, and service providers who need a lightweight, web-friendly, and machine-readable representation of IndoorGML datasets.



SECURITY CONSIDERATIONS

No security considerations have been made for this Standard.



SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- The University of New South Wales
- Pusan National University
- CityGeometrix
- National Institute of Advanced Industrial Science and Technology



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1

SCOPE

The OGC IndoorGML 2.0 Part 2b – JSON Encoding Standard defines a JSON encoding for IndoorGML 2.0 Part 1 – Conceptual Model. The encoding is referred to as IndoorJSON.

The IndoorJSON Standard specifies:

- JSON schema for IndoorGML 2.0 core module;
- JSON schema for IndoorGML 2.0 navigation module;
- the geometry encoding rules based on JSON-FG geometry objects; and
- a statement to which IndoorJSON conformance classes a IndoorGML 2.0 conforms to.

IndoorJSON is an implementation encoding of IndoorGML 2.0 Part 1. Therefore, IndoorJSON instances shall be interpreted according to the semantics defined in the IndoorGML 2.0 conceptual model.



2

CONFORMANCE

This standard defines implementation specification which specifies how the IndoorGML 2.0 Conceptual Model should be implemented using JavaScript Object Notation (JSON). The Standardization Target for this standard is:

- Implementations of the IndoorGML 2.0 Conceptual Model using JSON encodings.

Conformance with this standard shall be checked using all the relevant tests specified in Annex A (normative) of this document. The framework, concepts, and methodology for testing, and the criteria to be achieved to claim conformance are specified in the OGC Compliance Testing Policies and Procedures and the OGC Compliance Testing web site.

In order to conform to this OGC® interface standard, a software implementation shall choose to implement:

- Any one of the conformance levels specified in Annex A (normative).

All requirements-classes and conformance-classes described in this document are owned by the standard(s) identified.

Table 2

CONFORMANCE CLASS	URI
IndoorGML Core	http://www.opengis.net/spec/indoorgml-2/2.0/json/conf/core
Navigation Extension	http://www.opengis.net/spec/indoorgml-2/2.0/json/conf/navi



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NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Johannes Echterhoff: OGC 24-017r1, *Best Practice for OGC – UML to JSON Encoding Rules*. Open Geospatial Consortium (2025). <http://www.opengis.net/doc/BP/uml-to-json-encoding/1.0>.

Sisi Zlatanova, Abdoulaye Diakite, Taehoon Kim, Ki-Joune Li: OGC 22-045r5, *OGC IndoorGML 2.0 Part 1 – Conceptual Model*. Open Geospatial Consortium (2025). <http://www.opengis.net/doc/IS/indoorgml/2.0>.

Clemens Portele, Panagiotis (Peter) A. Vretanos: OGC 21-045r1, *OGC Features and Geometries JSON – Part 1: Core*. Open Geospatial Consortium (2026). <http://www.opengis.net/doc/IS/json-fg-1/1.0>.

ISO: ISO 19101-1, *Geographic information – Reference model – Part 1: Fundamentals*. International Organization for Standardization, Geneva <https://www.iso.org/standard/59164.html>.

ISO: ISO 19107, *Geographic information – Spatial schema*. International Organization for Standardization, Geneva <https://www.iso.org/standard/66175.html>.

T. Bray (ed.): IETF RFC 8259, *The JavaScript Object Notation (JSON) Data Interchange Format*. RFC Publisher (2017). <https://www.rfc-editor.org/info/rfc8259>.



4

TERMS AND DEFINITIONS

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the ‘ModSpec’. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

4.1. conceptual model

model that defines concepts of a universe of discourse

[SOURCE: ISO 19101-1]

4.2. conceptual schema

formal description of a conceptual model

[SOURCE: ISO 19101-1]

4.3. feature

abstraction of real world phenomena

Note 1 to entry: Features are objects that have an identity that allows for distinguishing features from each other. Features can have spatial and non-spatial properties that describe the features in more detail. Spatial properties are properties that have a geometry from ISO 19107 as data type.

Note 2 to entry: Features are denoted by the stereotype «FeatureType» in the IndoorGML 2.0 Conceptual Model.

[SOURCE: ISO 19101-1]



5

CONVENTIONS

5.1. Identifiers

The normative provisions in this standard are denoted by the URI

<http://www.opengis.net/spec/indoorgml-2/2.0/json>

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

5.2. Use of JSON terminology

The following terms are used as specified in JSON. In particular:

- “member” refers to a name/value pair of an object;
- “key” refers to the name of a member; and
- “value” refers to the value of a member, which may be an object, array, string, number, boolean, or null.

For example, the GeoJSON “geometry” member has the key “geometry” and one of the GeoJSON geometries (or null) as its value.



6

OVERVIEW OF INDOORJSON

IndoorJSON implements the conceptual classes of IndoorGML 2.0 Part 1 as JSON objects. The conceptual model remains normative for semantics, while this document defines the JSON representation.

6.1. Design Principle

IndoorJSON is based on the following encoding principles.

6.1.1. Conceptual model preservation

IndoorJSON Core preserve the IndoorGML 2.0 core conceptual model structure, as shown in the below figure.

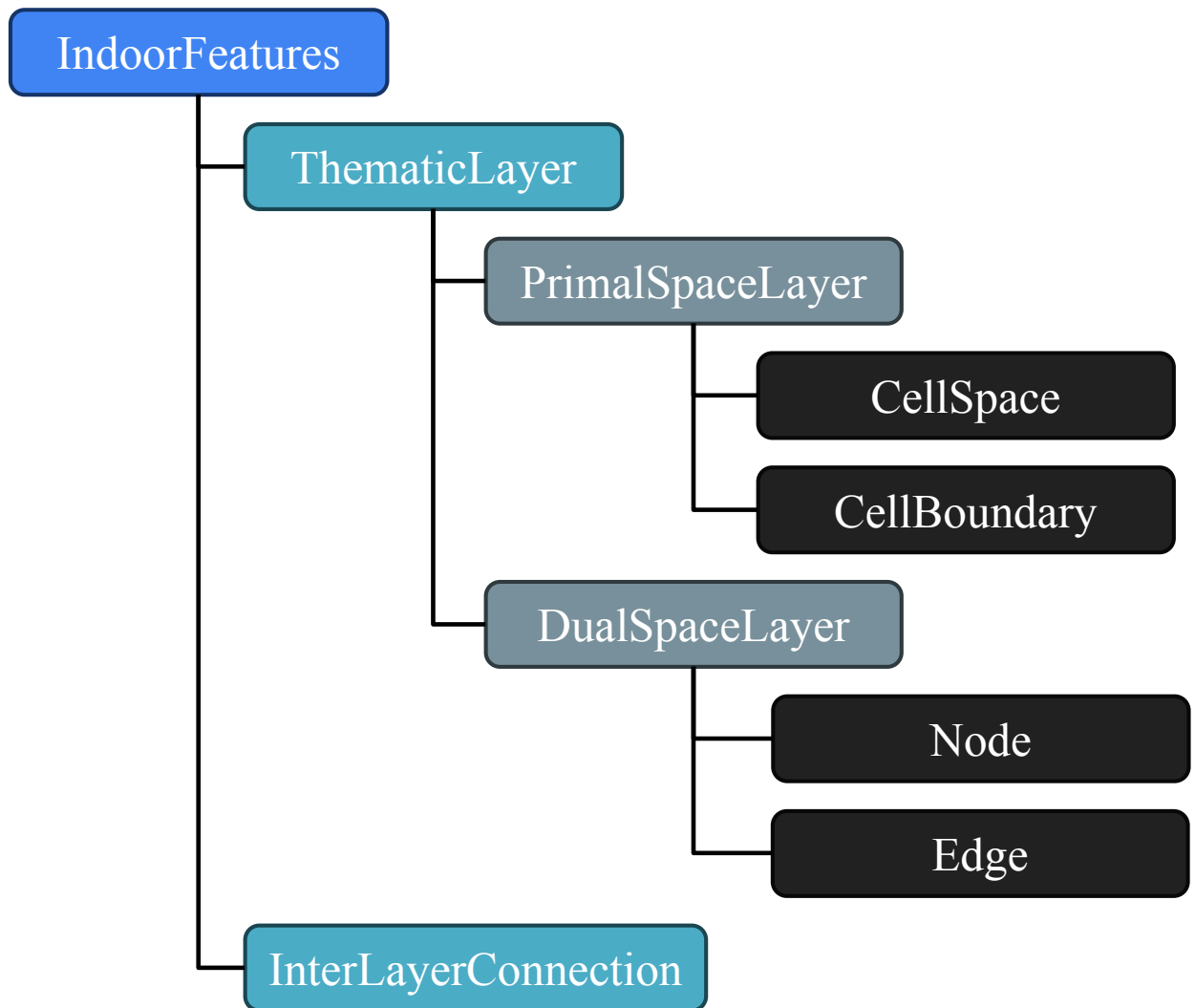


Figure 1 – IndoorGML 2.0 core module structure

The IndoorJSON Core schema follows Figure 1. Each class is defined individually, and depending on their relationships, they can hold other classes as property values or array values. For example, **IndoorFeatures** have `layers` property for **ThematicLayer** and `layerConnections` property for **InterLayerConnection**.

IndoorJSON Navigation preserve the IndoorGML 2.0 navigation conceptual model structure, as shown in the below figure.

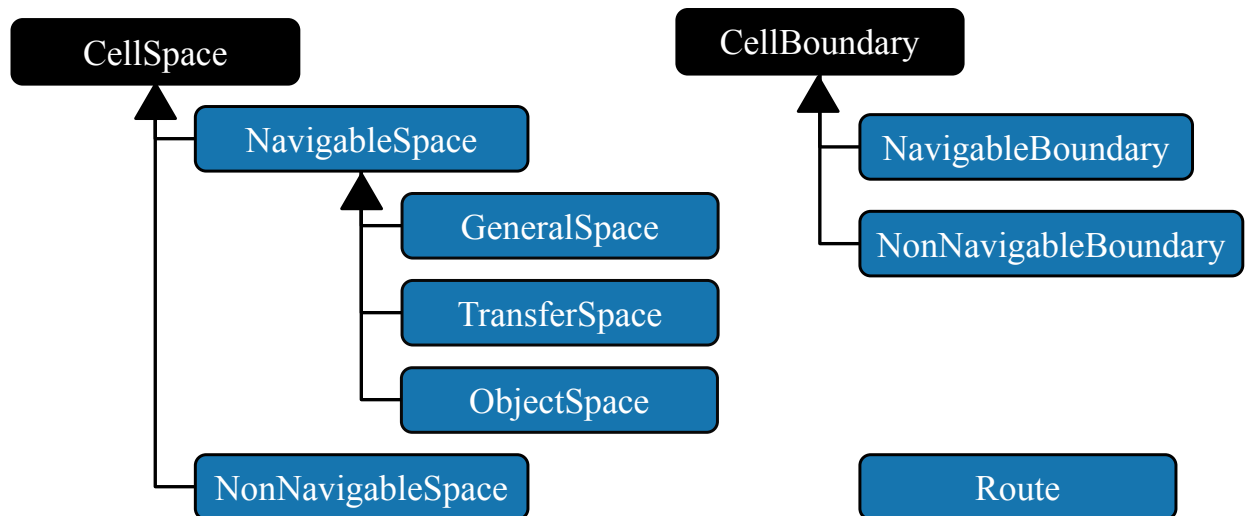


Figure 2 – IndoorGML 2.0 navigation module structure

The IndoorJSON Navigation schema follows Figure 1. In the figure, the arrows represent inheritance. Each class is defined individually. Depending on the inheritance relationship, a child class inherits the properties of its parent class.

6.1.2. JSON-FG-compatible geometry

IndoorJSON use JSON-FG-compatible geometry objects for the geometry representation. This is necessary because indoor spaces require 3D geometry, including solid geometry. The current IndoorJSON core schema references JSON-FG geometry schemas for `geometry2D` and `geometry3D`, including **Polygon** and **Polyhedron** for `CellSpace`.

7

INDOORJSON CORE

REQUIREMENTS CLASS 1

IDENTIFIER	http://www.opengis.net/spec/indoorgml-2/2.0/json/req/core
TARGET TYPE	Implementation Specification
PREREQUISITES	http://www.opengis.net/spec/indoorgml/2.0/req http://www.opengis.net/spec/json-fg-1/1.0/req/core http://www.opengis.net/spec/json-fg-1/1.0/req/polyhedra
NORMATIVE STATEMENTS	Requirement 1: /req/core/indoorjson Requirement 1-7: /req/core/dualspace Requirement 2: /req/core/indoorfeature Requirement 3: /req/core/thematiclayer Requirement 4: /req/core/primalspacelayer Requirement 5: /req/core/cellspace Requirement 6: /req/core/cellboundary Requirement 8: /req/core/node Requirement 9: /req/core/edge Requirement 10: /req/core/interlayerconnection

IndoorJSON object represents one of feature class of IndoorGML 2.0 conceptual model.

The IndoorJSON Object:

- **must** have one member with the name `id`. The value must be a string that serves as a unique identifier for the IndoorJSON object and can be used to represent cross-references between IndoorJSON objects.
- **must** have one member with the name `featuretype`. The value is one of the possibilities in the figure Figure 1.

REQUIREMENT 1: REQUIREMENT FOR INDOORJSON OBJECT

IDENTIFIER	/req/core/indoorjson
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/json/req/core
A	An IndoorJSON object SHALL contain <code>id</code> and <code>featuretype</code> members.

7.1. IndoorFeatures

IndoorFeature object is an instance of IndoorFeature, the container for the indoor dataset. It aggregates one or more thematic layers and may include inter-layer connections.

The **IndoorFeature** object shall contain:

- id
- featureType
- layers for representing **ThematicLayer** object as item
- optionally, layerConnections for **InterLayerConnection** object as item

```
{
  "IndoorFeatures": {
    "$anchor": "IndoorFeatures",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["IndoorFeatures"] },
      "layers": {
        "type": "array",
        "minItems": 1,
        "items": { "$ref": "#/$defs/ThematicLayer" },
        "uniqueItems": true
      },
      "layerConnections": {
        "type": "array",
        "items": { "$ref": "#/$defs/InterLayerConnection" },
        "uniqueItems": true
      }
    },
    "required": ["id", "featureType", "layers"]
  }
}
```

Listing 1 – A JSON schema of the IndoorFeature:

```
{
  "featureType": "IndoorFeatures",
  "layers": [
    {
      "id": "TL-1",
      "featureType": "ThematicLayer",
      "semanticExtension": false,
      "theme": "Physical",
      "primalSpace": {...},
      "dualSpace": {...}
    },
    ...
  ],
  "layerConnections": [
    {
      "id": "ILC-1",
      "featureType": "InterLayerConnection",

```

```

    "connectedLayers": ["TL-1", "TL-2"],
    "connectedNodes": ["N1", "N101"]
  }
]
}

```

Listing 2 — An example of the IndoorFeature object:

REQUIREMENT 2: INDOORFEATURES SCHEMA	
IDENTIFIER	/req/core/indoorfeature
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/json/req/core
A	An IndoorFeatures object SHALL contain at least one ThematicLayer object as item of layers value.

7.2. ThematicLayer

ThematicLayer object is an instance of ThematicLayer, which represents one contextual decomposition of indoor space. For example, physical layer, navigation layer, sensor layer, etc.

The **ThematicLayer** object shall contain:

- id
- featureType
- semanticExtension indicates whether there is extension module with additional semantic information associated to the **PrimalSpaceLayer**
- theme determines what type of model representation can be expected in the **PrimalSpaceLayer**
- optionally, primalSpace for **PrimalSpaceLayer** object
- optionally, dualSpace for **DualSpaceLayer** object

The core schema defines semanticExtension and theme members, with theme values including "Virtual", "Physical", "Tags", and "Unknown", according to the conceptual model.

```

{
  "ThematicLayer": {
    "$anchor": "ThematicLayer",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["ThematicLayer"] },
      "semanticExtension": { "type": "boolean" },

```

```

    "theme": { "enum": ["Virtual", "Physical", "Tags", "Unknown"] },
    "primalSpace": { "$ref": "#/$defs/PrimalSpaceLayer" },
    "dualSpace": { "$ref": "#/$defs/DualSpaceLayer" }
  },
  "required": ["id", "featureType", "semanticExtension", "theme"]
}

```

Listing 3 — A JSON schema of the ThematicLayer:

```

{
  "id": "TL-1",
  "featureType": "ThematicLayer",
  "semanticExtension": false,
  "theme": "Physical",
  "primalSpace": {...},
  "dualSpace": {...}
}

```

Listing 4 — An example of the ThematicLayer object:

REQUIREMENT 3: THEMATICLAYER SCHEMA

IDENTIFIER	/req/core/thematiclayer
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/json/req/core
A	A ThematicLayer object SHALL declare whether there is an extension module with additional semantic information using the semanticExtension member.
B	A ThematicLayer object SHALL determine what type of model representation can be expected using the theme member.
C	The Theme value SHALL be set to “Virtual”, “Physical”, “Tags”, and “Unknown”.

7.3. PrimalSpaceLayer

PrimalSpaceLayer object is an instance of PrimalSpaceLayer, which contains spatial objects representing indoor subdivisions.

The **PrimalSpaceLayer** object shall contain:

- id
- featureType
- cellSpaceMember for representing **CellSpace** object as item
- optionally, cellBoundaryMember for representing **CellBoundary** object as item

- optionally, creationDatetime
- optionally, terminationDatetime

```
{
  "PrimalSpaceLayer": {
    "$anchor": "PrimalSpaceLayer",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["PrimalSpaceLayer"] },
      "creationDatetime": { "type": "string", "format": "date-time" },
      "terminationDatetime": { "type": "string", "format": "date-time" },
      "cellSpaceMember": {
        "type": "array",
        "minItems": 1,
        "items": { "$ref": "#/$defs/CellSpace" },
        "uniqueItems": true
      },
      "cellBoundaryMember": {
        "type": "array",
        "items": { "$ref": "#/$defs/CellBoundary" },
        "uniqueItems": true
      }
    },
    "required": ["id", "featureType", "cellSpaceMember"]
  }
}
```

Listing 5 — A JSON schema of the PrimalSpaceLayer:

```
{
  "primalSpace": {
    "id": "PS-1",
    "featureType": "PrimalSpaceLayer",
    "creationDatetime": "2025-10-30T13:00:00",
    "cellSpaceMember": [...],
    "cellBoundaryMember": [...]
  }
}
```

Listing 6 — An example of the PrimalSpaceLayer object:

REQUIREMENT 4: PRIMALSPACE LAYER SCHEMA

IDENTIFIER	/req/core/primalspacelayer
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorqml-2/2.0/json/req/core
A	A PrimalSpaceLayer SHALL contain one or more CellSpace objects.

7.4. CellSpace

CellSpace object is an instance of `CellSpace`, which represents a unit of indoor space. The **CellSpace** may correspond to a room, corridor, stairwell, elevator cabin, platform, zone, virtual partition, or other spatial subdivision.

An **CellSpace** object shall contain:

- `id`
- `featureType`
- `poi` is a flag that specifies whether this **CellSpace** is a point of interest
- optionally, `cellSpaceName` is the name assigned to the space according to internal convention
- optionally, `level`
- optionally, `duality` can be mapped to the `id` value of the **Node** object
- optionally, `cellSpaceGeom`
- optionally, `boundedBy` can have the `id` value of the linked **CellBoundary** objects
- optionally, `externalReference` for representing **ExternalReference** object

A **CellSpace** can be linked to a **Node** through the `duality` member. A **CellSpace** can be linked to a set of **CellBoundaries** through the `boundedBy` member. For `cellSpaceGeom` member, it defines `cellSpaceGeom.geometry2D` using a JSON-FG Polygon geometry and `cellSpaceGeom.geometry3D` using a JSON-FG Polyhedron geometry. The `externalReference` member is used for the reference of an object to its corresponding object in an external data set.

```
{
  "CellSpace": {
    "$anchor": "CellSpace",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["CellSpace"] },
      "cellSpaceName": { "type": "string" },
      "level": { "type": "string" },
      "poi": { "type": "boolean" },
      "duality": { "type": "string", "format": "uri-reference" },
      "cellSpaceGeom": {
        "type": "object",
        "properties": {
          "geometry2D": { "$ref": "https://beta.schemas.opengis.net/json-fg/geometry-object.json#/$defs/Polygon" },
          "geometry3D": { "$ref": "https://beta.schemas.opengis.net/json-fg/geometry-object.json#/$defs/Polyhedron" }
        }
      },
      "boundedBy": {
```

```

        "type": "array",
        "items": { "type": "string", "format": "uri-reference" },
        "uniqueItems": true
    },
    "externalReference": { "$ref": "#/$defs/ExternalReferenceType" }
},
"required": ["id", "featureType", "poi"]
}
}

```

Listing 7 — A JSON schema of the CellSpace:

```

{
  "id": "C1",
  "featureType": "CellSpace",
  "duality": "TL-1:DS-1:N1",
  "poi": false,
  "level": "1",
  "cellSpaceGeom": {
    "geometry2D": {
      "type": "Polygon",
      "coordinates": [[[0,0], [2,0], [2,2], [0,2], [0,0]]]
    }
  },
  "boundedBy": ["TL-1:PS-1:B1"]
}

```

Listing 8 — An example of the CellSpace object:

REQUIREMENT 5: CELLSPACE SCHEMA

IDENTIFIER	/req/core/cellspace
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorqml-2/2.0/json/req/core
A	A CellSpace object SHALL declare whether it is a point of interest space using the <code>poi</code> member.

7.5. CellBoundary

CellBoundary object is an instance of CellBoundary, a boundary of a CellSpace. A boundary may be physical or virtual.

The **CellBoundary** object shall contain:

- `id`
- `featureType`
- `isVirtual` indicate whether a **CellBoundary** corresponds to a virtual surface (true) or a physical one (false)

- optionally, duality can be mapped to the id value of the **Edge** object
- optionally, cellBoundaryGeom
- optionally, externalReference for representing **ExternalReference** object

A **CellBoundary** can be linked to a **Edge** through the duality member. For cellBoundaryGeom member, it defines cellBoundaryGeom.geometry2D using a JSON-FG **LineString** geometry and cellBoundaryGeom.geometry3D using a JSON-FG **Polygon** geometry. The externalReference member is used for the reference of an object to its corresponding object in an external data set

```
{
  "CellBoundary": {
    "$anchor": "CellBoundary",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["CellBoundary"] },
      "isVirtual": { "type": "boolean" },
      "duality": { "type": "string", "format": "uri-reference" },
      "cellBoundaryGeom": {
        "type": "object",
        "properties": {
          "geometry2D": { "$ref": "https://beta.schemas.opengis.net/json-fg/geometry-object.json#/$defs/LineString" },
          "geometry3D": { "$ref": "https://beta.schemas.opengis.net/json-fg/geometry-object.json#/$defs/Polygon" }
        }
      },
      "externalReference": { "$ref": "#/$defs/ExternalReferenceType" }
    },
    "required": ["id", "featureType", "isVirtual"]
  }
}
```

Listing 9 – A JSON schema of the CellBoundary:

```
{
  "id": "B1",
  "featureType": "CellBoundary",
  "isVirtual": false,
  "duality": "TL-1:DS-1:E1",
  "cellBoundaryGeom": {
    "geometry2D": {
      "type": "LineString",
      "coordinates": [[2,0],[2,2]]
    }
  }
}
```

Listing 10 – An example of the CellBoundary object:

REQUIREMENT 6: CELLBOUNDARY SCHEMA

IDENTIFIER	/req/core/cellboundary
------------	------------------------

REQUIREMENT 6: CELLBOUNDARY SCHEMA

INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/json/req/core
A	A CellBoundary object SHALL specify whether it is virtual using the <code>isVirtual</code> member.

7.6. DualSpaceLayer

DualSpaceLayer object is an instance of DualSpaceLayer, represents the graph structure corresponding to the primal space layer. It contains nodes and edges.

The **DualSpaceLayer** object shall contain:

- `id`
- `featureType`
- `isLogical` indicates whether the provided graph is a geometric or a logical
- `isDirected` specifies whether the provided graph is directed
- `nodeMember` for representing **Node** object as item
- optionally, `edgeMember` for representing **Edge** object as item
- optionally, `creationDatetime`
- optionally, `terminationDatetime`

```
{
  "DualSpaceLayer": {
    "$anchor": "DualSpaceLayer",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["DualSpaceLayer"] },
      "creationDatetime": { "type": "string", "format": "date-time" },
      "terminationDatetime": { "type": "string", "format": "date-time" },
      "isLogical": { "type": "boolean" },
      "isDirected": { "type": "boolean" },
      "nodeMember": {
        "type": "array",
        "minItems": 1,
        "items": { "$ref": "#/$defs/Node" },
        "uniqueItems": true
      },
      "edgeMember": {
        "type": "array",
        "items": { "$ref": "#/$defs/Edge" },
        "uniqueItems": true
      }
    }
  }
}
```

```

    },
    "required": ["id", "featureType", "isLogical", "isDirected", "nodeMember"]
  }
}

```

Listing 11 — A JSON schema of the DualSpaceLayer:

```

{
  "dualSpace": {
    "id": "DS-1",
    "featureType": "DualSpaceLayer",
    "creationDatetime": "2025-10-30T13:00:00",
    "isLogical": false,
    "isDirected": false,
    "nodeMember": [...],
    "edgeMember": [...]
  }
}

```

Listing 12 — An example of the DualSpaceLayer object:

REQUIREMENT 7: DUALSPACE LAYER SCHEMA

IDENTIFIER	/req/core/dualspace layer
A	A DualSpaceLayer object SHALL specify whether the provided graph is a geometric or a logical using the <code>isLogical</code> member.
B	A DualSpaceLayer object SHALL specify whether the provided graph is directed using the <code>isDirected</code> member.
C	A DualSpaceLayer object SHALL contain one or more Node objects.

7.7. Node

Node object is an instance of [Node](#), a state or a graph vertex in the dual space layer.

The **Node** object shall contain:

- `id`
- `featureType`
- `duality` must be mapped to the `id` member of the **CellSpace** object
- optionally, `geometry`
- optionally, `connects` can have the `id` member of the connected **Edge** object

A **Node** may be linked to a **CellSpace** through the duality member. A **Node** may be connects to a set of **Edges** through the connects member. For geometry member referred JSON-FG Point geometry.

```
{
  "Node": {
    "$anchor": "Node",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["Node"] },
      "duality": { "type": "string", "format": "uri-reference" },
      "geometry": { "$ref": "https://beta.schemas.opengis.net/json-fg/geometry-object.json#/$defs/Point" },
      "connects": {
        "type": "array",
        "items": { "type": "string", "format": "uri-reference" },
        "uniqueItems": true
      }
    },
    "required": ["id", "featureType", "duality"]
  }
}
```

Listing 13 – A JSON schema of the Node:

```
{
  "id": "N1",
  "featureType": "Node",
  "duality": "TL-1:PS-1:C1",
  "geometry": {
    "type": "Point",
    "coordinates": [1,1]
  },
  "connects": ["TL-1:DS-1:E1"]
}
```

Listing 14 – An example of the Node object:

REQUIREMENT 8: NODE SCHEMA	
IDENTIFIER	/req/core/node
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/json/req/core
A	A Node object SHALL contain a duality value by referencing the id member of to correspond CellSpace .

7.8. Edge

Edge object is an instance of `Edge`, which represents a relationship between two nodes. It may represent adjacency, connectivity, accessibility, or navigability.

The Edge object shall contain:

- `id`
- `featureType`
- `weight` can be used in graph-based applications
- `connects` must have the `id` value of the connected **Nodes** object
- optionally, `duality` can be mapped to the `id` value of the **CellBoundary** object
- optionally, `geometry`

An Edge must reference the connected **Nodes** using the `connects` member. An Edge can be linked to a **CellBoundary** through the `duality` member. For `geometry` member referred JSON-FG `LineString` geometry.

```
{
  "Edge": {
    "$anchor": "Edge",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["Edge"] },
      "duality": { "type": "string", "format": "uri-reference" },
      "weight": { "type": "number" },
      "geometry": { "$ref": "https://beta.schemas.opengis.net/json-fg/geometry-object.json#/$defs/LineString" },
      "connects": {
        "type": "array",
        "items": { "type": "string", "format": "uri-reference" },
        "minItems": 2,
        "maximum": 2,
        "uniqueItems": true
      }
    },
    "required": ["id", "featureType", "weight", "connects"]
  }
}
```

Listing 15 — A JSON schema of an Edge:

```
{
  "id": "E1",
  "featureType": "Edge",
  "duality": "B1",
  "weight": 1.0,
  "geometry": {
    "type": "LineString",
    "coordinates": [[1,1],[3,1]]
  }
}
```

```

    },
    "connects": [
      "TL-1:DS-1:N1",
      "TL-1:DS-1:N2"
    ]
  }
}

```

Listing 16 — An example of the Edge object:

REQUIREMENT 9: EDGE SCHEMA

IDENTIFIER	/req/core/edge
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/json/req/core
A	An Edge object SHALL reference the connected Nodes using the connects member.
B	An Edge object SHALL contain a weight value.

7.9. InterLayerConnection

InterLayerConnection object is an instance of [InterLayerConnection](#), represents a relationship between objects in different thematic layers.

The **InterLayerConnection** object shall contain:

- id
- featureType
- connectedLayers must be mapped to the id value of the two **ThematicLayer** objects
- typeOfTopoExpression
- optionally, connectedNodes can be mapped to the id value of the two **Node** objects
- optionally, connectedCells can be mapped to the id value of the two **CellSpace** objects
- optionally, comment

The typeOfTopoExpression member represents the topological relationship between two layers. The typeOfTopoExpression values including "CONTAINS", "OVERLAPS", "EQUALS", "WITHIN", "CROSSES", and "OTHERS", according to the conceptual model. Those topological values are in the form of verbs for which the subject is the first instance of the connectedLayers member. The values of the connectedLayers, connectedNodes, and connectedCells members are ordered and can be mapped to the same index.

```

{
  "InterLayerConnection": {
    "$anchor": "InterLayerConnection",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["InterLayerConnection"] },
      "connectedLayers": {
        "type": "array",
        "items": { "type": "string", "format": "uri-reference" },
        "minItems": 2,
        "maximum": 2,
        "uniqueItems": true
      },
      "connectedNodes": {
        "type": "array",
        "items": { "type": "string", "format": "uri-reference" },
        "minItems": 2,
        "maximum": 2,
        "uniqueItems": true
      },
      "connectedCells": {
        "type": "array",
        "items": { "type": "string", "format": "uri-reference" },
        "minItems": 2,
        "maximum": 2,
        "uniqueItems": true
      },
      "typeOfTopoExpression": {
        "enum": ["CONTAINS", "OVERLAPS", "EQUALS", "WITHIN", "CROSSES", "OTHERS"]
      },
      "comment": { "type": "string" }
    },
    "required": ["id", "featureType", "connectedLayers", "typeOfTopoExpression"]
  }
}

```

Listing 17 – A JSON schema of the InterLayerConnection:

```

{
  "id": "ILC-1",
  "featureType": "InterLayerConnection",
  "connectedLayers": ["TL-1", "TL-2"],
  "connectedNodes": ["N1", "N101"]
}

```

Listing 18 – An example of the InterLayerConnection object:

REQUIREMENT 10: INTERLAYERCONNECTION SCHEMA

IDENTIFIER	/req/core/interlayerconnection
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/json/req/core

7.10. ExternalReferenceType

The `externalReferenceType` object is used for the reference of an object to its corresponding object in an external data set. This is not a feature class of IndoorGML 2.0.

```
{
  "ExternalReferenceType": {
    "$anchor": "ExternalReferenceType",
    "type": "object",
    "properties": {
      "externalObject": {
        "type": "object",
        "properties": {
          "name": { "type": "string" },
          "uri": { "type": "string", "format": "uri" }
        },
        "required": ["name", "uri"]
      },
      "informationSystem": { "type": "string", "format": "uri" }
    },
    "required": ["externalObject", "informationSystem"]
  }
}
```

Listing 19 — A JSON schema of the `ExternalReferenceType`:



8

INDOORJSON NAVIGATION

REQUIREMENTS CLASS 2

IDENTIFIER	http://www.opengis.net/spec/indoorgml-2/2.0/json/req/navi
TARGET TYPE	Implementation Specification
PREREQUISITES	http://www.opengis.net/spec/indoorgml/2.0/req http://www.opengis.net/spec/json-fg-1/1.0/req/core http://www.opengis.net/spec/json-fg-1/1.0/req/polyhedra
NORMATIVE STATEMENTS	Requirement 2-1: /req/navi/navigablespace Requirement 2-2: /req/navi/generalspace Requirement 2-3: /req/navi/transferspace Requirement 2-4: /req/navi/objectspace Requirement 2-5: /req/navi/navigableboundary Requirement 2-6: /req/navi/route



ANNEX A (NORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE



ANNEX A

(NORMATIVE)

CONFORMANCE CLASS ABSTRACT TEST SUITE

NOTE: Ensure that there is a conformance class for each requirements class and a test for each requirement (identified by requirement name and number)

A.1. Conformance Class A

Example

REQUIREMENT A.1

STATEMENT	conf/service/profile/basic-wps
-----------	--------------------------------

RECOMMENDATION A.1

STATEMENT	Verify that the server implements the Basic WPS conformance class.
-----------	--

REQUIREMENT A.2

STATEMENT	Verify that the server implements the Synchronous WPS and/or the Asynchronous WPS conformance class. Verify that the requests and responses to a supported operation are syntactically correct. Verify that the service supports the Synchronous WPS Conformance class, the Asynchronous WPS Conformance class or both. Verify that all process offerings implement the native process model. Verify the following list of conformance tests: • Conformance test 1 • Conformance test 2
-----------	--



ANNEX B (NORMATIVE) SCHEMAS

B

ANNEX B (NORMATIVE) SCHEMAS

B.1. JSON Schema of a IndoorJSON Core

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://www.opengis.net/spec/indoorgml/2.0/json/indoorjson.core.schema.json",
  "title": "IndoorJSON for IndoorGML 2.0 core part",
  "description": "IndoorJSON Core",
  "oneOf": [
    { "$ref": "#/$defs/IndoorFeatures" },
    { "$ref": "#/$defs/ThematicLayer" },
    { "$ref": "#/$defs/InterLayerConnection" },
    { "$ref": "#/$defs/PrimalSpaceLayer" },
    { "$ref": "#/$defs/CellSpace" },
    { "$ref": "#/$defs/CellBoundary" },
    { "$ref": "#/$defs/DualSpaceLayer" },
    { "$ref": "#/$defs/Node" },
    { "$ref": "#/$defs/Edge" }
  ],
  "$defs": {
    "IndoorFeatures": {
      "$anchor": "IndoorFeatures",
      "type": "object",
      "properties": {
        "id": { "type": "string" },
        "featureType": { "enum": ["IndoorFeatures"] },
        "layers": {
          "type": "array",
          "minItems": 1,
          "items": { "$ref": "#/$defs/ThematicLayer" },
          "uniqueItems": true
        },
        "layerConnections": {
          "type": "array",
          "items": { "$ref": "#/$defs/InterLayerConnection" },
          "uniqueItems": true
        }
      },
      "required": ["id", "featureType", "layers"]
    },
    "ThematicLayer": {
      "$anchor": "ThematicLayer",

```

```

    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["ThematicLayer"] },
      "semanticExtension": { "type": "boolean" },
      "theme": { "enum": ["Virtual", "Physical", "Tags", "Unknown"] },
      "primalSpace": { "$ref": "#/$defs/PrimalSpaceLayer" },
      "dualSpace": { "$ref": "#/$defs/DualSpaceLayer" }
    },
    "required": ["id", "featureType", "semanticExtension", "theme"]
  },
  "PrimalSpaceLayer": {
    "$anchor": "PrimalSpaceLayer",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["PrimalSpaceLayer"] },
      "creationDatetime": { "type": "string", "format": "date-time" },
      "terminationDatetime": { "type": "string", "format": "date-time" },
      "cellSpaceMember": {
        "type": "array",
        "minItems": 1,
        "items": { "$ref": "#/$defs/CellSpace" },
        "uniqueItems": true
      },
      "cellBoundaryMember": {
        "type": "array",
        "items": { "$ref": "#/$defs/CellBoundary" },
        "uniqueItems": true
      }
    },
    "required": ["id", "featureType", "cellSpaceMember"]
  },
  "CellSpace": {
    "$anchor": "CellSpace",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["CellSpace"] },
      "cellSpaceName": { "type": "string" },
      "level": { "type": "string" },
      "poi": { "type": "boolean" },
      "duality": { "type": "string", "format": "uri-reference" },
      "cellSpaceGeom": {
        "type": "object",
        "properties": {
          "geometry2D": { "$ref": "https://beta.schemas.opengis.net/json-fg/geometry-object.json#/$defs/Polygon" },
          "geometry3D": { "$ref": "https://beta.schemas.opengis.net/json-fg/geometry-object.json#/$defs/Polyhedron" }
        }
      },
      "boundedBy": {
        "type": "array",
        "items": { "type": "string", "format": "uri-reference" },
        "uniqueItems": true
      },
      "externalReference": { "$ref": "#/$defs/ExternalReferenceType" }
    },
    "required": ["id", "featureType", "poi"]
  },

```

```

"CellBoundary": {
  "$anchor": "CellBoundary",
  "type": "object",
  "properties": {
    "id": { "type": "string" },
    "featureType": { "enum": ["CellBoundary"] },
    "isVirtual": { "type": "boolean" },
    "duality": { "type": "string", "format": "uri-reference" },
    "cellBoundaryGeom": {
      "type": "object",
      "properties": {
        "geometry2D": { "$ref": "https://beta.schemas.opengis.net/json-fg/
geometry-object.json#/$defs/LineString" },
        "geometry3D": { "$ref": "https://beta.schemas.opengis.net/json-fg/
geometry-object.json#/$defs/Polygon" }
      }
    },
    "externalReference": { "$ref": "#/$defs/ExternalReferenceType" }
  },
  "required": ["id", "featureType", "isVirtual"]
},

"DualSpaceLayer": {
  "$anchor": "DualSpaceLayer",
  "type": "object",
  "properties": {
    "id": { "type": "string" },
    "featureType": { "enum": ["DualSpaceLayer"] },
    "creationDatetime": { "type": "string", "format": "date-time" },
    "terminationDatetime": { "type": "string", "format": "date-time" },
    "isLogical": { "type": "boolean" },
    "isDirected": { "type": "boolean" },
    "nodeMember": {
      "type": "array",
      "minItems": 1,
      "items": { "$ref": "#/$defs/Node" },
      "uniqueItems": true
    },
    "edgeMember": {
      "type": "array",
      "items": { "$ref": "#/$defs/Edge" },
      "uniqueItems": true
    }
  },
  "required": ["id", "featureType", "isLogical", "isDirected", "nodeMember"]
},

"Node": {
  "$anchor": "Node",
  "type": "object",
  "properties": {
    "id": { "type": "string" },
    "featureType": { "enum": ["Node"] },
    "duality": { "type": "string", "format": "uri-reference" },
    "geometry": { "$ref": "https://beta.schemas.opengis.net/json-fg/
geometry-object.json#/$defs/Point" },
    "connects": {
      "type": "array",
      "items": { "type": "string", "format": "uri-reference" },
      "uniqueItems": true
    }
  }
},

```



```

    "required": ["id", "featureType", "duality"]
  },
  "Edge": {
    "$anchor": "Edge",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["Edge"] },
      "duality": { "type": "string", "format": "uri-reference" },
      "weight": { "type": "number" },
      "geometry": { "$ref": "https://beta.schemas.opengis.net/json-fg/geometry-object.json#/$defs/LineString" },
      "connects": {
        "type": "array",
        "items": { "type": "string", "format": "uri-reference" },
        "minItems": 2,
        "maximum": 2,
        "uniqueItems": true
      }
    },
    "required": ["id", "featureType", "weight", "connects"]
  },
  "InterLayerConnection": {
    "$anchor": "InterLayerConnection",
    "type": "object",
    "properties": {
      "id": { "type": "string" },
      "featureType": { "enum": ["InterLayerConnection"] },
      "connectedLayers": {
        "type": "array",
        "items": { "type": "string", "format": "uri-reference" },
        "minItems": 2,
        "maximum": 2,
        "uniqueItems": true
      },
      "connectedNodes": {
        "type": "array",
        "items": { "type": "string", "format": "uri-reference" },
        "minItems": 2,
        "maximum": 2,
        "uniqueItems": true
      },
      "connectedCells": {
        "type": "array",
        "items": { "type": "string", "format": "uri-reference" },
        "minItems": 2,
        "maximum": 2,
        "uniqueItems": true
      },
      "typeOfTopoExpression": {
        "enum": ["CONTAINS", "OVERLAPS", "EQUALS", "WITHIN", "CROSSES",
"OTHERS"]
      },
      "comment": { "type": "string" }
    },
    "required": ["id", "featureType", "connectedLayers", "typeOfTopoExpression"]
  },
  "ExternalReferenceType": {
    "$anchor": "ExternalReferenceType",
    "type": "object",

```

```

    "properties": {
      "externalObject": {
        "type": "object",
        "properties": {
          "name": { "type": "string" },
          "uri": { "type": "string", "format": "uri" }
        },
        "required": ["name", "uri"]
      },
      "informationSystem": { "type": "string", "format": "uri" }
    },
    "required": ["externalObject", "informationSystem"]
  }
}

```

Listing B.1

B.2. JSON Schema of a IndoorJSON Navigation

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://www.opengis.net/spec/indoorgml/2.0/json/indoorjson.navi.schema.json",
  "title": "IndoorJSON for IndoorGML 2.0 navigation part",
  "description": "IndoorJSON Navigation",
  "oneOf": [
    { "$ref": "#/$defs/NavigableSpace" },
    { "$ref": "#/$defs/NonNavigableSpace" },
    { "$ref": "#/$defs/GeneralSpace" },
    { "$ref": "#/$defs/TransferSpace" },
    { "$ref": "#/$defs/ObjectSpace" },
    { "$ref": "#/$defs/NavigableBoundary" },
    { "$ref": "#/$defs/NonNavigableBoundary" },
    { "$ref": "#/$defs/Route" }
  ],
  "$defs": {
    "NavigableSpace": {
      "$anchor": "NavigableSpace",
      "allOf": [
        { "$ref": "indoorjson.core.schema.json#/$defs/CellSpace" },
        {
          "type": "object",
          "properties": {
            "locomotionType": { "enum": ["Flying", "Rolling", "Walking", "Unspecified"] }
          },
          "required": ["locomotionType"]
        }
      ]
    },
    "NonNavigableSpace": {
      "$anchor": "NonNavigableSpace",
      "allOf": [
        { "$ref": "indoorjson.core.schema.json#/$defs/CellSpace" }
      ]
    }
  }
}

```

```

    ]
  },
  "GeneralSpace": {
    "$anchor": "GeneralSpace",
    "allOf": [
      { "$ref": "#/$defs/NavigableSpace" },
      {
        "type": "object",
        "properties": {
          "function": { "enum": ["Administration", "Business, trade",
"Education, training", "Recreation", "Laboratory", "Storage", "Security"]}
        },
        "required": ["function"]
      }
    ]
  },
  "TransferSpace": {
    "$anchor": "TransferSpace",
    "allOf": [
      { "$ref": "#/$defs/NavigableSpace" },
      {
        "type": "object",
        "properties": {
          "category": { "enum": ["Door", "Window"] },
          "function": { "enum": ["AnchorSpace", "ConnectionSpace"] }
        },
        "required": ["category", "function"]
      }
    ]
  },
  "ObjectSpace": {
    "$anchor": "ObjectSpace",
    "allOf": [
      { "$ref": "#/$defs/NonNavigableSpace" },
      {
        "type": "object",
        "properties": {
          "containedFeatures": { "type": "integer" },
          "description": { "type": "string" }
        }
      }
    ]
  },
  "NavigableBoundary": {
    "$anchor": "NavigableBoundary",
    "allOf": [
      { "$ref": "indoorjson.core.schema.json#/$defs/CellBoundary" },
      {
        "type": "object",
        "properties": {
          "boundaryOrientation": { "type": "boolean" },
          "navigableBoundaryFunction": { "enum": ["AnchorBoundary",
"ConnectionBoundary"]}
        },
        "required": ["navigableBoundaryFunction"]
      }
    ]
  },

```

```

"NonNavigableBoundary": {
  "$anchor": "NonNavigableBoundary",
  "allOf": [
    {"$ref": "indoorjson.core.schema.json#/$defs/CellBoundary"}
  ]
},

"Route": {
  "$anchor": "Route",
  "type": "object",
  "properties": {
    "creationDate": { "type": "string", "format": "date-time" },
    "routeNode": {
      "type": "array",
      "minItems": 2,
      "items": {"$ref": "indoorjson.core.schema.json#/$defs/Node"}
    },
    "routeEdge": {
      "type": "array",
      "minItems": 1,
      "items": {"$ref": "indoorjson.core.schema.json#/$defs/Edge"}
    }
  },
  "required": ["routeNode", "routeEdge"]
}
}
}

```

Listing B.2



ANNEX C (INFORMATIVE) REVISION HISTORY



ANNEX C (INFORMATIVE) REVISION HISTORY

DATE	RELEASE	EDITOR	PRIMARY CLAUSES MODIFIED	DESCRIPTION
2026-06-03	0.1	Taehoon Kim	all	initial version



BIBLIOGRAPHY





BIBLIOGRAPHY

NOTE: The TC has approved Springer LNCS as the official document citation type.

Springer LNCS is widely used in technical and computer science journals and other publications

For citations in the text please use square brackets and consecutive numbers: [1], [2], [3]

Actual References:

[n] Journal: Author Surname, A.: Title. Publication Title. Volume number, Issue number, Pages Used (Year Published)

- [1] ISO: ISO 19142, *Geographic information – Web Feature Service*. International Organization for Standardization, Geneva <https://www.iso.org/standard/42136.html>.
- [2] W3C: **Data Catalog Vocabulary**, W3C Recommendation 16 January 2014, <https://www.w3.org/TR/vocab-dcat/>
- [3] IANA: **Link Relation Types**, <https://www.iana.org/assignments/link-relations/link-relations.xml>
- [4] W3C/OGC: **Spatial Data on the Web Best Practices**, W3C Working Group Note 28 September 2017, <https://www.w3.org/TR/sdw-bp/>
- [5] W3C: **Data on the Web Best Practices**, W3C Recommendation 31 January 2017, <https://www.w3.org/TR/dwbp/>
- [6] Ben-Kiki, O., Evans, C., Ingy döt Net: **YAML Ain't Markup Language**, <https://yaml.org/>
- [7] OGC: **Web Feature Service 2.0**, <http://docs.opengeospatial.org/is/09-025r2/09-025r2.html>
- [8] Berners-Lee, T., Fielding, R., Masinter, L.: **IETF RFC 3986 – Uniform Resource Identifier (URI): Generic Syntax**, <http://tools.ietf.org/rfc/rfc3986.txt>